

Homework 2: Cart Pole Optimal Control

SES 598: Space Robotics and AI

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Problem Statement:

The objective of this assignment is to use a linear quadratic regulator (LQR) controller to optimally control and stabilize a cart-pole system under randomized earthquake-like disturbances, while keeping the cart within 2.5 meters of its starting position. The cart pole system is a simple underactuated robot that consists of a cart on a track with an inverted pendulum mounted on the top. The cart must move horizontally on the track in a controlled in order to keep the pole in the upright position.

An LQR controller is an optimal control technique that minimizes the following quadratic cost function, which balances the state deviation and control input.

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

Where:

- x is the state vector $[x, \dot{x}, \theta, \dot{\theta}]^T$
- u is the control input (horizontal force on the cart)
- Q is the state cost matrix, which penalizes deviations from the desired state
- R is the control effort cost, which penalizes large force inputs

System Constraints:

- Cart position limit: $\pm 2.5 \text{ m}$
- Control rate: 50 Hz
- Cart mass: 1.0 kg
- Pole mass: 1.0 kg
- Pole length: 1.0 m

Earthquake-like Disturbances:

- Base amplitude: 15 N
- Frequency range: $0.5\text{-}4.0 \text{ Hz}$
- Random variations in amplitude and phase
- Additional Gaussian noise

Baseline Performance:

The performance of the controller was analyzed using these stability metrics:

- Duration of stable operation: t_{max}
- Maximum pole angle deviation: $\theta_{max} \equiv \max(|\theta|)$

- Maximum cart displacement: $x_{max} \equiv \max(|x|)$
- Average control effort: $u_{avg} \equiv \text{mean}(|u|)$

These metrics were combined into a stability score using

$$\text{stability score} = \max(0, 10 - 2 \cdot x_{max} - 0.2 \cdot \theta_{max} - 0.05 \cdot u_{avg})$$

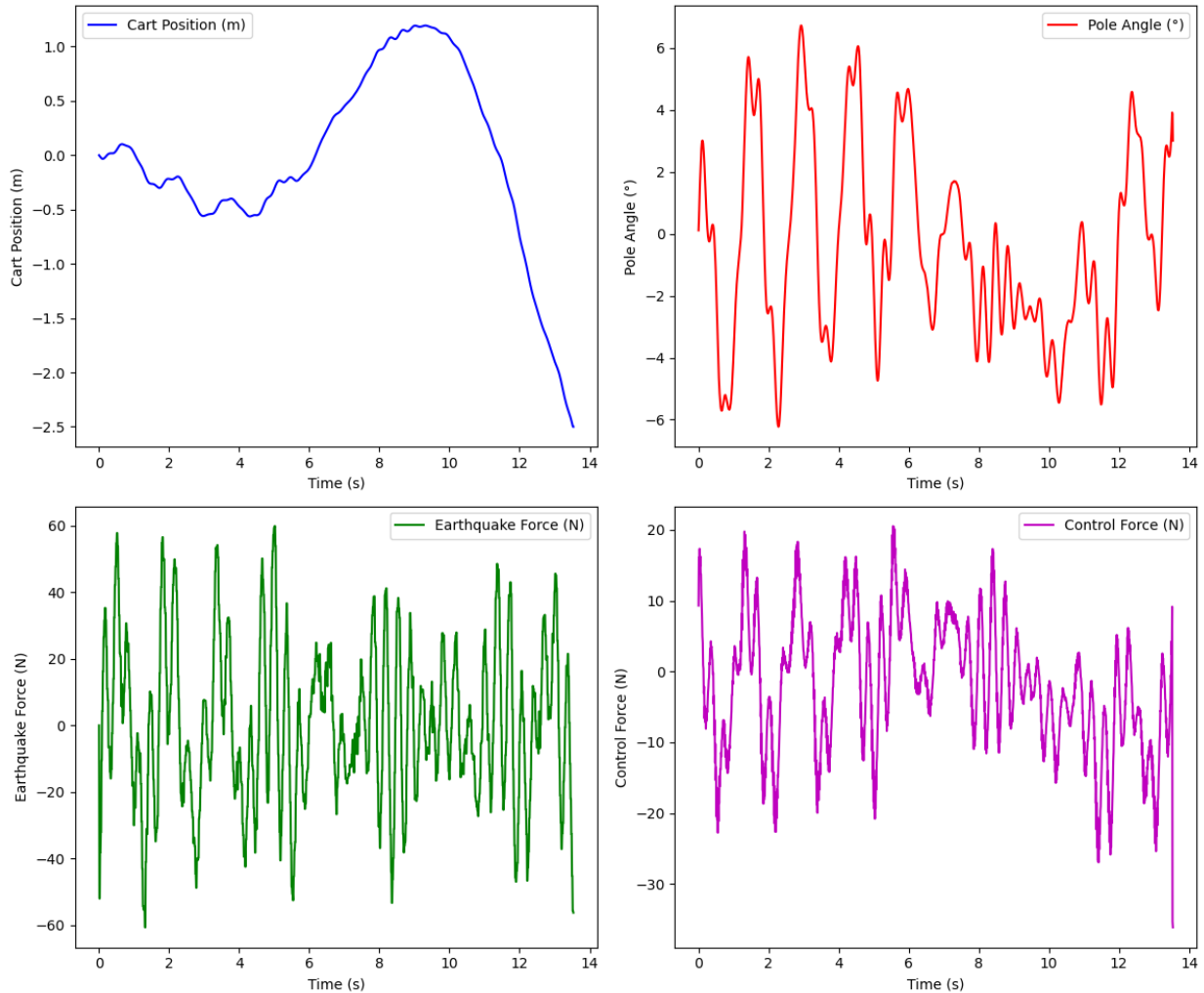
The sim has a time limit of 120 seconds which ensures stability over long periods of time.

The baseline performance was tested with two following cost matrices:

$$Q = [1.0, 1.0, 10.0, 10.0]$$

$$R = [0.1]$$

This resulted in:



These plots show that base performance of the controller is not great. While it is able to maintain a fairly small pole angle, the cart position gets out of control fairly quickly, remaining the desired area of operation for a little over 13 seconds. This shows that the controller needs to be tuned in a way that prioritizes the position more. Between 8 and 11 seconds the angle of the was held close to -2° which caused the pole to start to hold a drift in the negative direction to counteract it. This

shows that the controller should ideally keep the angle oscillating around 0° without prolonged time with either a positive or negative angle.

The performance metrics for the baseline test are as follows:

- $t_{\max} = 13.54 \text{ s}$
- $\theta_{\max} = 6.720^\circ$
- $x_{\max} = 2.500 \text{ m}$
- $u_{\text{avg}} = 7.375 \text{ N}$
- $\text{stability score} = 3.29$

Tuning the LQR

First the Q matrix was varied with a constant R in order to find a more optimal controller. The Q matrix was varied first since the R value is related to the efficiency of the control input and finding a better Q will lead to more stable states. For these tests R was held at its default $R = 0.1$. For ease of testing, the Q values associated with each velocity ($\dot{x}$ and $\dot{\theta}$) were kept the same as their corresponding position state (x and θ).

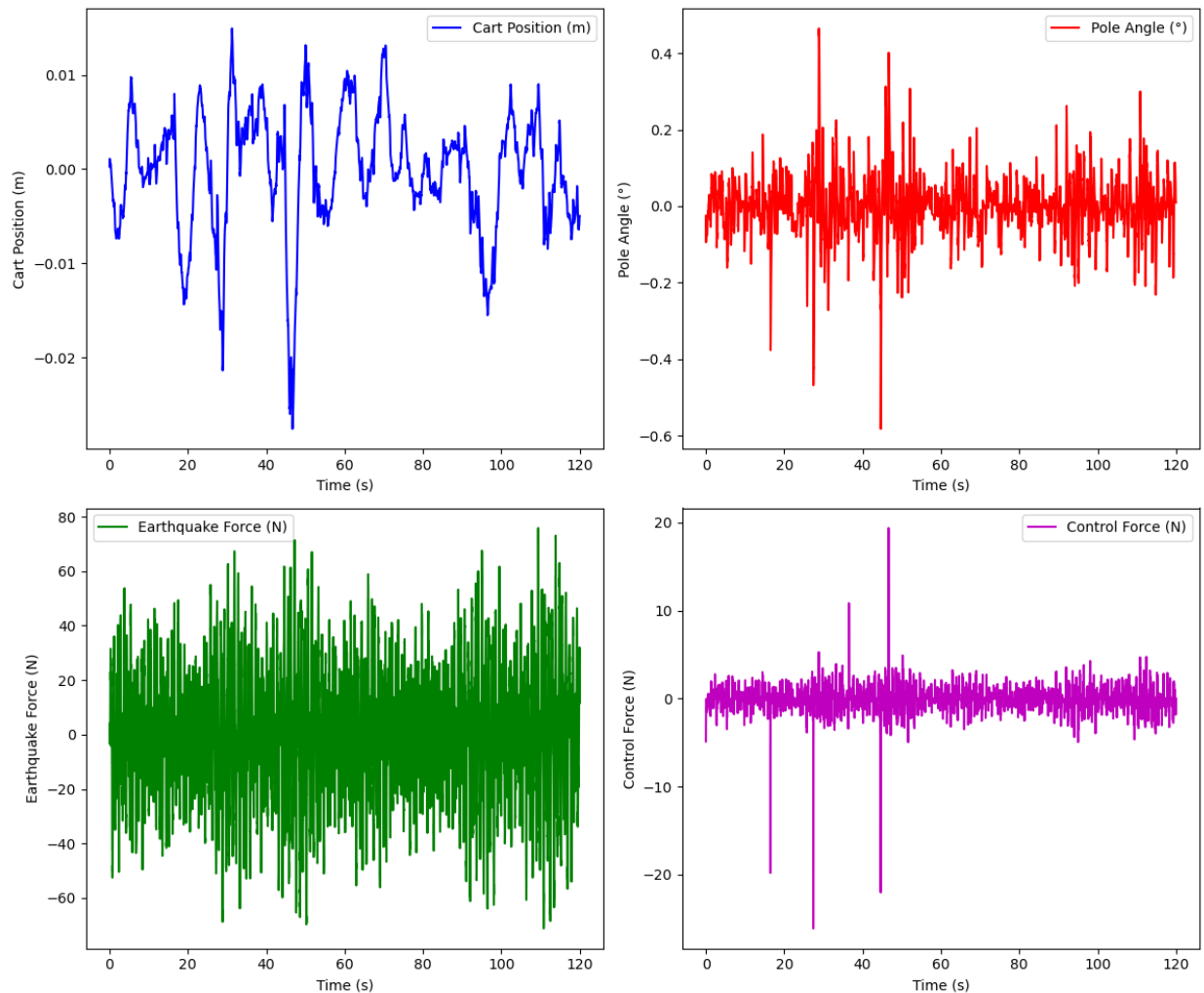
<i>Q Values</i>	$t_{\max} \text{ (s)}$	$x_{\max} \text{ (m)}$	$\theta_{\max} \text{ (}^\circ\text{)}$	$u_{\text{avg}} \text{ (N)}$	<i>Stability Score</i>
[10,10,10,10]	120	0.077	3.375	5.730	8.89
[10,10,50,50]	120	0.022	1.178	0.723	9.75
[20,20,20,20]	120	0.024	1.299	3.216	9.53
[5,5,25,25]	120	0.182	3.325	5.840	8.70
[10,10,30,30]	120	0.076	1.956	3.270	9.29
[20,20,50,50]	120	0.027	0.936	2.399	9.64
[10,10,25,25]	120	0.044	1.191	3.331	9.51

Then a more optimal value of R was determined by $Q = [10,10,50,50]$, and trying a small variety of different R values.

<i>R Values</i>	$t_{\max} \text{ (s)}$	$x_{\max} \text{ (m)}$	$\theta_{\max} \text{ (}^\circ\text{)}$	$u_{\text{avg}} \text{ (N)}$	<i>Stability Score</i>
0.01	120	0.016	0.604	6.348	9.53
0.25	120	0.108	3.232	4.021	8.94
0.13	120	0.122	3.003	6.993	8.80
0.05	120	0.025	0.582	0.333	9.81

Results and Discussion:

The optimal controller found for this cart-pole setup had $Q = [10, 10, 50, 50]$ and $R = [0.05]$. This setup resulted in:



The plots above show that both the cart's and the pole's positions do not stray far from the equilibrium point. This helps keep the input force down when compared to the baseline performance. With a few outlying spikes in force which are approximately the same as the max force on the baseline controller. It was observed in the baseline performance, that the state cost had to increase. Both the cart and the pole varied too much from their desired positions, which led to the controller failing to keep the cart in bounds. The decrease in R allowed for more aggressive controls that could keep up with the disturbances.

The tuning method used was once again tedious similar to the PD controller for the turtle sim, but it felt easier to see the effects that each change made on the overall performance of the system. Overall tuning the LQR controller was fairly simple, once it was observed what changing the state costs did to the performance.

